Subject Chimists Date 30/9/2018 No. sunday

البنية الذرية — خصائص الدورية للعناصر

* Atomic structure and periodic properties

universe (الكون) → matter (مادة) → pure (نقي) ⟹ 1. Element مثل Li, Na, K ...
matter → impure (غير نقي) → 2. compaond مثل H_2O, NH_3, ...
universe → Energy (طاقة)

Element: It is a pure substance (مادة نقية) formed from same Type of atoms.
compaund: It is a pure substance formed from two or more types of atoms (الذرات)

The atom (الذرات) is formed from nucleons (نوات) and electrons (الإلكترونات). The nucleons is formed from protons (البروتون), which are tinny particls bearing a possitive $^{+}$ charge and neutrons $^{+-}$, which have also a tinny particles bearing neutral charges. The electrons are very tinny particles bearing negative $^{-}$ charge and distributing (يتوزع) around the nucleous in certain energy levels

energy levels (مستوى الطاقة) = shell (n)
* shell 1 = 1 subshell (s)
* shell 2 = 2 subshell (s, p)
* shell 3 = 3 subshell (s, p, d)
* shell 4 = 4 subshell (s, p, d, f)

	No. of e's ($2n^2$)
s → 1 orbital	2 e's
p → 3 orbital (P_x, P_y, P_z)	6 e's
d → 5 - orbital ($d_{xy}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{z^2}$)	10 e's
f → 7 - orbital	14 e's
g → 9 - orbital	18 e's

NOTEBOOK

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Quantum Numbers (Four numbers)

1. principle Quantum Number (n): It gives information about the shell number in which the electron exists. $n = 1, 2, 3 \ldots$ (for azmuthal QN)

2. secondary Quantum No (L): It gives information about the subshell in which the electron exists. $L = s, p, d, f, g, h, i \ldots$
$L = 0, 1, 2, 3, 4, 5 \ldots (n-1)$

3. Magnetic Quantum No (m_L): It gives information about the orbital in which the electron exists. $(m_L) = -L \ldots, 0, \ldots +L$

* for p-subshell: $m_L = -1, 0, +1$. (P_x, P_y, P_z)
* s-subshell: $m_L(s) = L(s) = 0$.
* d-subshell: $L(d) = 2 \Rightarrow m_L = -2, -1, 0, 1, 2$. ($d_{xy}$ d_{xz} d_{yz} $d_{x^2-y^2}$ d_{z^2})
* f-subshell: $L(f) = 3 \Rightarrow m_L = -3, -2, -1, 0, 1, 2, 3$

4. spin Quantum No (m_s): It gives information about the spinning direction. $m_s = +\frac{1}{2}, -\frac{1}{2}$

* set of Quantum Numbers :- (n, L, m_L, m_s)

Example: $(\underset{\text{shell}}{4}, \underset{s}{0}, \underset{\text{عدد الكم المغزلي}}{0}, -\frac{1}{2}) \rightarrow 4s$

Example(2): which of the following sets of Q.N is/are reasonable and which is not? why?

1. $(5, 4, 4, -\frac{1}{2})$
2. $(6, 8, -2, \pm\frac{1}{2})$
3. $(4, 1, -2, +\frac{1}{2})$
4. $(3, 3, -1, 1)$

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*the number of orbitals in any subshell is equal to $2l+1$:

number of p-orbitals = $2*1+1=3$

" " d- " = $2*2+1=5$

" " f- " = $2*3+1=7$

*Rules govern the electronic configuration. (Build-up or Aufbau principle)

[1] fill the shell with lowest n-value first. (energy = $l+n$)

[2] In the same shell, fill the subshell with lowest l-value first.

*example: which one should be filled first: 5s or 4d? 5s

$n+l=5+0=5$ $4+2=6$

~~$n+l=(5s)=5+0=5$~~

$n+l=(4d)=4+2=6$

[3] In higher shells, the energy is govern by $n+l$ one has to fill the subshell with lowest $(n+l)$ value first. If the value is The same, fill the subshell with lowest n-value first. $(n+l)$

example: which one should be filled first 5f or 7P?

$n+l=(5f)=5+3=8$

$n+l=(7P)=7+1=8$

} 5f filled first.

[4] By filling a subshell with multi-orbitals, fill the electrons one-by-one, such that the number of unpaired electrons is max and they should have same spinning direction. (Hunds Rule)

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Examples:- $_7N: 1s^2\,2s^2\,2p^3$ (1↑ 1↑ 1↑) — يوجد 3 الكترونات مفردة فيه

$_{16}S: 1s^2\,2s^2\,2p^6\,3s^2\,3p^4$ (3s: 1↑; 3p: 1↑ 1↑ 1) — يوجد الكترونين مفردين فيه

5) The atom gains more stability, when it ended with full or half-full d or f-subshell, such that, if needs one electron only to reach this state, it will take it from the nearest s-orbital

Ex: $_{24}Cr: 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^4$ (1↑ 1↑ 1↑ 1↑) ~~1 1 1 1~~

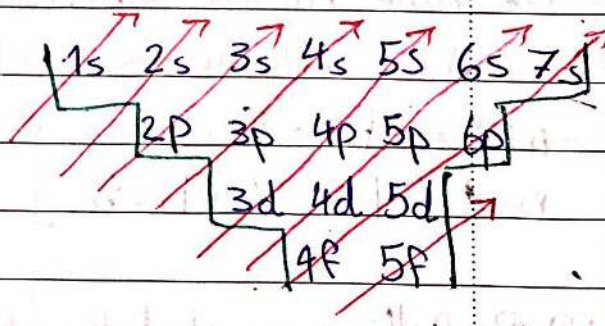
التوزيع الصحيح

$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^1\,3d^5$ (1↑ 1↑ 1↑ 1↑ 1↑)

* We have 6-unpaired electron's

• $_{29}Cu: 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^1\,3d^{10}$

• $_{79}Au: 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^{10}\,4p^6\,5s^2\,4d^{10}\,5p^6\,6s^1\,4f^{14}\,5d^{10}$



• $_{33}As: 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^{10}\,4p^3$

$As = [_{18}Ar]\,4s^2\,3d^{10}\,4p^3$

Nobel gases (الغازات الخاملة): It's electronic configuration ended with np^6 except except $_2He: 1s^2$, $_{10}Ne: 1s^2\,2s^2\,2p^6$, $_{18}Ar$, $_{36}Kr$, $_{54}Xe$, $_{86}Rn$

X: $_{49}: 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^{10}\,4p^6\,5s^2\,4d^{10}\,5p^1$

$[Kr]\,5s^2\,4d^{10}\,5p^1$

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$_{49}In : 1s^2)\ 2s^2\ 2p^6)\ 3s^2\ 3p^6\ 3d^{10})\ 4s^2\ 4p^6\ 4d^{10})\ 5s^2\ 5p^1$

inner shells مستويات الطاقة الداخلية (core electrons) (1,2,3)

outer shells مستويات الطاقة الخارجية (valence shells) (4,5) الكترونات التكافؤ

(60) → Atomic mass number = Σ protons + neutrons (العدد الكتلي الذري)

$60 = 28 + \text{neutrons}$

$32 = \text{neutrons}$

(28) → atomic number = no. of protons العدد الذري

$_{11}Na : 1s^2)\ 2s^2\ 2p^6)\ 3s^1 \Rightarrow {}_{11}Na^{+} : 1s^2)\ 2s^2\ 2p^6$

sodium ion → ايون صوديوم

$_{20}Ca : 1s^2)\ 2s^2\ 2p^6)\ 3s^2\ 3p^6)\ 4s^2 \Rightarrow {}_{20}Ca^{2+} : 1s^2)\ 2s^2\ 2p^6)\ 3s^2\ 3p^6)$

$_{33}As^{-3} : 1s^2)\ 2s^2\ 2p^6)\ 3s^2\ 3p^6)\ 4s^2\ 3d^{10}\ 4p^6$

negative ione → ايون سالب

* periodic table and periodic properties (صانف الجدول الدوري) (خصائص الذرة)

① Family A :- electronic configuration ended with S or P (main group element)

② Family B :- The electronic configuration ended with D or F (transition metals)

if it ended with S : ① s^1 (Alkali metals العناصر القلوية) ② s^2 (Alkaline earth metals العناصر القلوية الترابية)

($_{87}Fr$ $_{55}Cs$ $_{3}Li$ $_{11}Na$ $_{11}K$ $_{37}Rb$) ($_{4}Be$ $_{12}Mg$ $_{20}Ca$ $_{38}Sr$ $_{56}Ba$ $_{88}Ra$)

if it ended with P : p^5 (halogens) $_{9}F$ $_{17}Cl$ $_{35}Br$ $_{53}I$ $_{85}At$)

p^6 (Nobel gases : ($_{2}He$ $_{10}Ne$ $_{18}Ar$ $_{36}Kr$ $_{54}Xe$ $_{86}Rn$)

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Subject Date 16th oct No. 6

Family B :- ended with D → maine transition metals (العناصر الانتقالية الرئيسية)

" "

ended with F → inner transition metals (العناصر الانتقالية الداخلية)

- 4f → Lanthanides (La)
- 5f → Actonides (Ac)

1: I
2: II
3: III
4: IV
5: V
6: VI
7: VII
8: VIII
9: IX
10: X

* The elements in periodic table are classified into groups (columns) and periods (rows)

In family A; the group no. is equal to the sum of e's in last (s+p). However, the period no. is equal to the hightest n-value.

Ex :- $_{35}Br$: $1s^2) 2s^2\ 2p^6) 3s^2\ 3p^6) 4s^2\ 3d^{10}\ 4p^5$ ⇒ Group VII A, Period 4

$_{12}Mg$: $1s^2\ 2s^2\ 2p^6\ 3s^2$ ⇒ Group II A, period 3

* In family B; the group No. is related to e's in (s+d) such that; (1) if Σ e's (s+d) ≤ 8 ⇒ the group No. is equal to the sum of e's.

(2) if Σ e's (s+d) = 8 or 9 or 10 ⇒ the group No is VIII B.

(3) If d is full ⇒ the group No. is equal to e's in s-only.

Ex :-

$_{25}Mn$: $1s^2) 2s^2\ 2p^6\ ) 3s^2\ 3p^6\ 4s^2\ 3d^5$ ⇒ Group VII B, period 4

$_{27}Co$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2 3d^7$ ⇒ Group VIII B, period 4

$_{26}Fe$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2 3d^6$ ⇒ Group VIII B, period 4

* period No. equal to the hightest n-value

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28 Ni : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$ ⇒ Group VIII B, Period 4

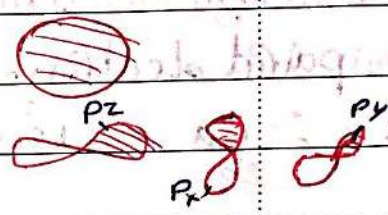
29 Cu : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$ ⇒ Group I B, period 4

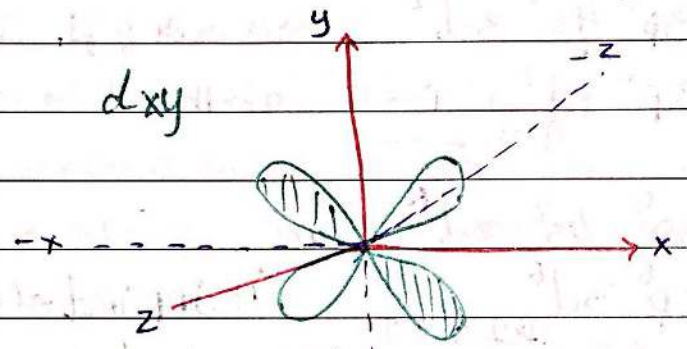
* Orbital shapes :-

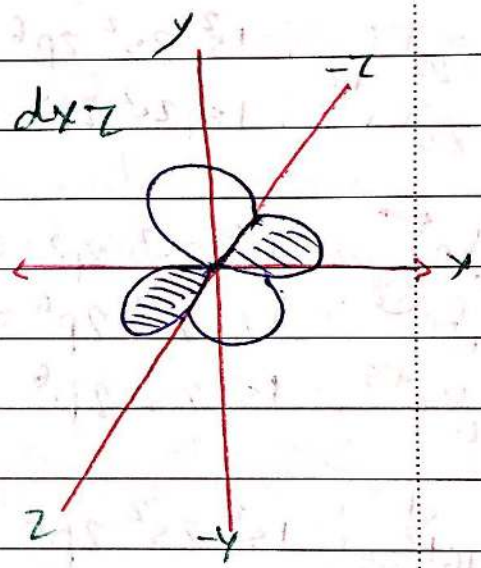
(1) S-orbitals :- sphirical كروي

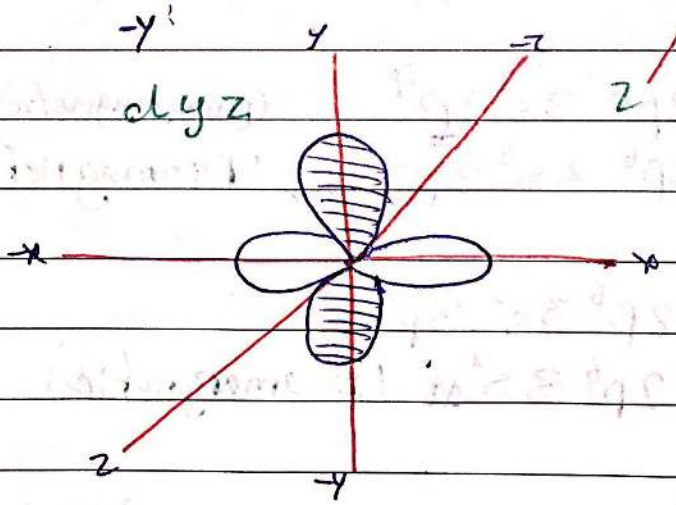
(2) p-orbitals :- dumbell دمبلي

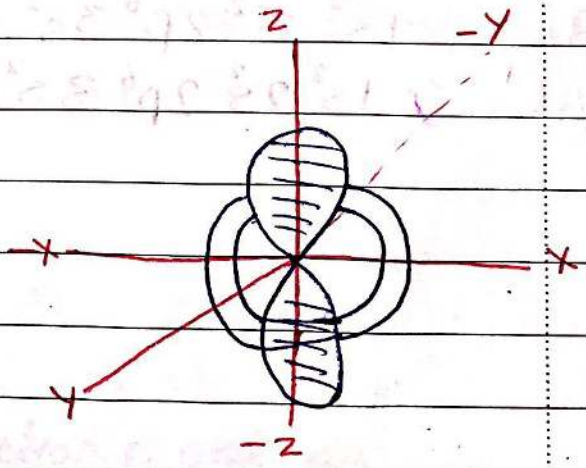
(3) d-orbital :- "cleaver leaf" ورقة البرسيم











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Subject Date No. 8

*paramagnetic element :- is the element which has at least one unpaired electron "It affects the magnetic field".

Ex: $_{74}W$ ⇒ paramagnetic . $_{8}O$: $1s^2 2s^2 2p^4$

*Diamagnetic element :- the element which hasn't any unpaired electron. It will not affect the magnetic field.

Ex :- $_{20}Ca$:- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ ↿⇂ Diamagnatic (Group IIA, IIB, VIII A)

* $_{22}Ti$:- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$

$_{22}Ti^{+2}$: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$ para

↿⇂ ↿ ⇀ ⇀

* $_{26}Fe^{+2}$:- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ para

$_{26}Fe^{+2}$:- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$

↿⇂ ↿ ↿ ↿ ↿ (4 unpaired els)

$_{26}Fe^{+3}$:- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ ↿ ↿ ↿ ↿ ↿ (5 unpaired els)

* $_{16}S$:- $1s^2 2s^2 2p^6 3s^2 3p^4$ (paramagnetic)

$_{16}S^{-2}$:- $1s^2 2s^2 2p^6 3s^2 3p^6$ ↿⇂ ↿⇂ ↿⇂ (Diamagnetic)

* $_{13}Al$:- $1s^2 2s^2 2p^6 3s^2 3p^1$

$_{13}Al^{+}$:- $1s^2 2s^2 2p^6 3s^2$ ↿⇂ (Diamagnetic)

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* periodic properties :- الخصائص الدورية

I. Atomic size :- الحجم الذري

$_3Li$:- $1s^2\ 2s^1$

$_{11}Na$:- $1s^2\ 2s^2\ 2p^6\ 3s^1$

$_{19}K$:- $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

$_{37}Rb$:- ..... $5s^1$

$_{55}Cs$:- ..... $6s^1$

$_{87}Fr$:- ..... $7s^1$

↓ يزداد الحجم الذري كل ما نزلنا لاسفل (Group IA)

$_4Be$:- $1s^2\ 2s^2$ ~~$2p^1$~~ / $_5B$:- $1s^2 2s^2 2p^1$ $_6C$:- $1s^2\ 2s^2\ 2p^2$

The atomic size increases يزداد from top to bottom in the same group, but decreases from left to right in the same period

* Ionic size : $_8O$: $1s^2\ 2s^2\ 2p^4$

$_8O^+$: $1s^2\ 2s^2\ 2p^3$

$_8O^{2+}$:- $1s^2\ 2s^2\ 2p^2$

$_8O^-$:- $1s^2\ 2s^2\ 2p^5$

$_8O^{-2}$:- $1s^2\ 2s^2\ 2p^6$

$_8O^{+2} < {}_8O^{+} < {}_8O < {}_8O^{-} < {}_8O^{-2}$

* possitively charged ions have smaller size than neutral atom. However negatively charged ions have larger size than the neutral atom.

* $_{13}Al^{+3}$: $1s^2\ 2s^2\ 2p^6$

$_{12}Mg^{+2}$: $1s^2\ 2s^2\ 2p^6$

$_{11}Na^{+}$: $1s^2\ 2s^2\ 2p^6$

$_{10}Ne$:- $1s^2\ 2s^2\ 2p^6$

$_7N^{-3}$: $1s^2\ 2s^2\ 2p^6$

$_7N^{-3} > {}_8O^{2-} > {}_9F^{-} > {}_{10}Ne > {}_{11}Na^{+} > {}_{12}Mg^{+2} > {}_{13}Al^{+3}$

Isoelectronic نفس العدد الإلكتروني

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(2) Ionization energy :- It is the min energy required to remove the farthest electron from isolated atom in its ground state and gaseous phase

$$X_{(g)} + \text{energy} \xrightarrow{\text{1st. I.E}} X^{+}_{(g)} + e^{-}$$

$$X^{+}_{(g)} + I_2 \xrightarrow{\text{2nd I.E}} X^{2+}_{(g)} + e^{-}$$

$$X^{2+}_{(g)} + I_3 \xrightarrow{\text{3rd I.E}} X^{3+}_{(g)} + e^{-}$$

$$I_3 > I_2 > I_1$$

$_3Li$:- $1s^2\ 2s^1$

$_{11}Na$:- $\ldots\ 3s^1$

$_{19}K$:- $\ldots\ 4s^1$

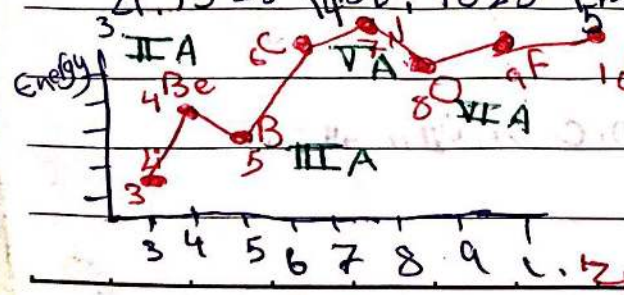
$_{37}Rb$:- $\ldots\ 5s^1$

$_{55}Cs$:- $\ldots\ 6s^1$

$_{87}Fr$:- $\ldots\ 7s^1$

تقل طاقة التأين كلما زاد الحجم تضعف قوة الارتباط

$_3Li: 1s^2 2s^1$ | $_4Be: 1s^2 2s^2$ | $_5B: 1s^2 2s^2 2p^1$ | ... | $_{10}Ne: 1s^2 2s^2 2p^6$



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Subject Date 23th oct No. 19

*the first Ionization energy (I1) decreases (↓) through the same group, how ever it increases (→) through the same period. but with some exceptions → between (II A and III A) and (V A and VI A)

*Electron Affinity :- It is the energy released, when an electron is added to an isolated atom in its ground state and gaseous phase

$$X_{(g)} + \bar{e} \longrightarrow X^-_{(g)} + \text{Energy}$$ "Electron Affinity" E.A

*the highest E.A is for group VII A and the lowest is for group VIII A (nobal gases)

electron Affinity :- Generally, most of elements in the periodic table tend to gain an electron to from negative ions except Group (VIII A nobal gases) and Group II A and N

$$X_{(g)} + \bar{e} \longrightarrow X^-_{(g)} + E.A$$ Exothermic process

(ΔH = -ve (negative)) ↳ the process is preferable

How ever, the ionization is endothermic process

(ΔH = +ve · possitive) ↳ unpreferable process

N O T E B O O K

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Subject Date No.

* The electron affinity is nearly stay constant Through same group, but it increases through the same period with exceptions (II A, VIII A)

* Ex:- Arrange the following elements according to increasing E.A ($_3Li$, $_8O$, $_7N$, $_{10}Ne$, $_5B$)

$$_{10}Ne < {}_7N < {}_3Li < {}_5B < {}_8O$$

* Nuclear effective charge:- (Z^*_{eff}) increases through the same period and slightly decreases (in ↓) the same group

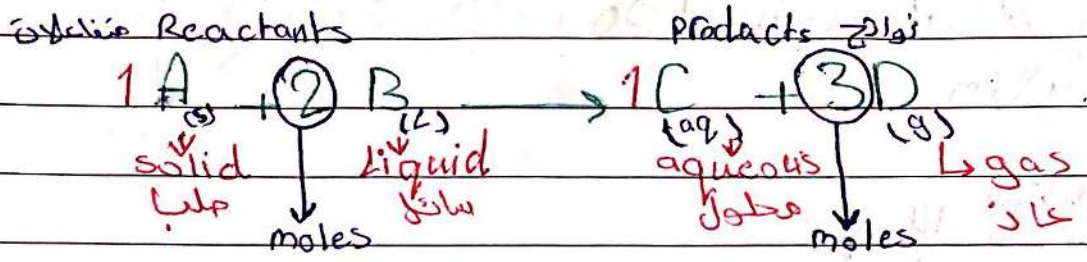
N O T E B O O K

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الحسابات الكيميائية

Stoichiometry

It means; chemical calculations based on balanced chemical equations.



The mole :- It is the amount of Carbon isotope-12, that exists in exactly 12g of this isotope which is equal to Avogadro's No (=6.22 x10^23 ^12_6C atom)

Average atomic mass :- It is the sum of atomic mass no. of each isotope multiplied by its natural abundance

molecular mass or molecular weight : (MWt)

It is the sum of atomic mass of each element multiplied by the number of each atom.

Ex :- what is the molecular mass of Glucose $C_6H_{12}O_6$ given that A.wt : (C = 12g/m, H = 1 g/m, O = 16g/m)

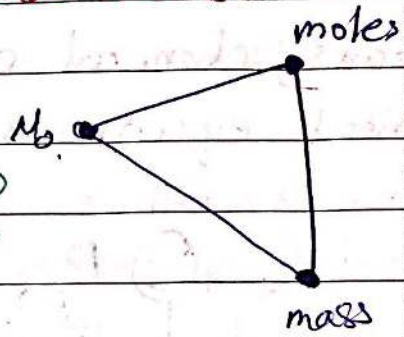
Atomic wight MWt = 6*12 g/m + 12*1 g/m + 6*16 g/m = 180g/m

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التحويل

Interconversion between moles, mass, and no. of atoms or molecules :-



Ex:- How many N-atoms in 480 g N?
A wt N = 14 g/mole

1 mole N weighting → 14 g
X?? → 480 g

$\frac{14X}{14} = \frac{480}{14} = 34.28$ mole N

1 mole of N contains → 6.022×10^{23} N-atoms
34.28 mole of N → ? Y

$Y = \frac{34.28 * 6.022 \times 10^{23} \text{ N-atoms}}{1 \text{ mole N}}$

$= 2.132 \times 10^{25}$ N-atoms

Ex:- what is the mass of 77 trillions of P-atoms?
(A wt P = 31 g/mole)

1 mole P contains → 6.022×10^{23} P-atoms
Y ? → 7.7×10^{13} P-atoms

$Y = \frac{7.7 \times 10^{13} \text{ P-atoms} * 1 \text{ mole P}}{6.022 \times 10^{23} \text{ P-atoms}}$

$= 1.28 \times 10^{-10}$ mole P

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Subject Date 30th oct No.

1 mole P $\xrightarrow{\text{weighting}}$ 31 g

1.28×10^{-10} mole P $\longrightarrow$? W

$W = 31\,g * \frac{1.28 \times 10^{-10} \text{ mole P}}{1} = 3.96 \times 10^{-9}$ g

(i.e. atoms ÷ Na)

Ex: How many molecules of CO_2 in 440 g CO_2?

Awt C = 12 g/mole, Awt O = 16 g/m.

m wt CO_2 = 1 × 12 g/mole + 2 × 16 g/m = 44 g/mole.

1 mole CO_2 $\xrightarrow{\text{weighting}}$ 44 g

Y ? $\xrightarrow{\times}$ 440 g

$Y = \frac{440 * 1 \text{ mole } CO_2}{44} = 10 \text{ mole } CO_2$

1 mole CO_2 $\xrightarrow{\text{contains}}$ $6.022 * 10^{23}$ molecule CO_2

10 mole CO_2 $\xrightarrow{\times}$ X

$X = 6.022 * 10^{24}$ molecule

Ex: what is the mass of Glucose ($C_6H_{12}O_6$) that contains 10^{28} C-atoms?

1 mole $C_6H_{12}O_6$ $\xrightarrow{\text{contains}}$ 6 C-atoms

W ? $\longrightarrow$ 10^{28} C-atoms

$W = 10^{28} \text{ C-atoms} * \frac{1 \text{ molecule}}{6 \text{ C-atoms}} = 1.67 * 10^{27}$ molecule

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Subject Date No.

1 mole $C_6H_{12}O_6$ contains $\rightarrow 6.022 \times 10^{23}$ molecule

Z_4 ? $\rightarrow 1.67 \times 10^{27}$ molecule

$$Z_4 = \frac{1 \text{ mole} \times 1.67 \times 10^{27}}{6.022 \times 10^{23}} = 2.67 \times 10^3 \text{ mole}$$

$$\text{mass} = 2.67 \times 10^3 \text{ mole} * \frac{180 \text{ g}}{\text{mole}} = 4.861 \times 10^5 \text{ g}$$

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Ex :- How many H-atoms in 46 g of benzoic acid $C_7H_6O_2$? (Hint: AWt : C = 12 g/mole, O = 16 g\mole H = 1 g\mole)

Molecular Mass = $7 \times 12 + 6 \times 1 + 2 \times 16 = 122$ g\mole.

* moles = $\frac{wt}{MWt}$ (الكتلة / الكتلة الجزيئية)

الوزن

$wt = moles * MWt$

$moles = \frac{Wt}{MWt} = \frac{4.6\ g}{122\ g/mole} = 0.038$ Mole

1 mole $C_7H_6O_2$ contains $\rightarrow 6.022 \times 10^{23}$ molecule

0.377 mole $\rightarrow$?? ×

$X = \frac{0.0377\ mole * 6.022 \times 10^{23}\ molecule}{1\ mole}$

$X = 2.27 \times 10^{22}$ molecule

1 molecule contains $\rightarrow$ 6 H-atoms

$2.27 \times 10^{22} \rightarrow$? ×

$Y = \frac{2.27 \times 10^{22}\ molecule \times 6\ H\text{-atoms}}{1\ molecule}$

$= 1.362 \times 10^{23}$ H-atoms

NOTEBOOK

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Subject Date 1st Nov No.

Mass percentage (m/m% or w/w%)
$= \frac{\text{wt of component}}{\text{total wt}} \times 100\%$

Ex: in benzoic acid, find the w/w% of C, O, H? $C_7H_6O_2$

$w/w\%\ C = \frac{\text{wt of C}}{\text{MWt}} \times 100\% = \frac{7 * 12\ g/mole}{122\ g/mole} \times 100\%$

$= 68.85\%$

$w/w\%\ H = \frac{6 \times 1\ g/mole}{122\ g/mole} \times 100\% = 4.92\%$

$w/w\%\ O = \frac{2 * 16}{122} \times 100\% = 26.23\%$

Ex(2): what is the mass percentage of water in $Na\,AL(SO_4)_2 \cdot 12H_2O$?

MWt $(NaAL(SO_4) \cdot 12H_2O)$

AWt: Na = 23 g/mole, AL = 27 g/mole, S = 32 g/mole
O = 16 g/mole, H = 1 g/mole.

$MWt = [23 \times 1 + 27 \times 1 + 2[32 \times 1 + 4 \times 16] + 12[2 \times 1 + 1 \times 16]]\ g/mole$
$= 462\ g/mole.$

$w/w\%\ H_2O = \frac{\text{wt of } H_2O}{MWt} \times 100\% = \frac{216}{462} \times 100\% = 46.53\%$

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Subject Date 1st Nov No.

Empirical formula : It is the simplest molar ratio between atoms forming certain compound

emp. form.

$C_6H_6 \Rightarrow CH$

molecular formula

Glucose emp. form.

$C_6H_{12}O_6 \Rightarrow CH_2O$

molecular formula

molecular formula :- It is the actual molar ratio between atoms forming certain compound.

Molecular formula = n * Emp. formula

Integer > 0

$n = \frac{\text{molecular Mass}}{\text{Emp. formula mass}}$

example :- A compound formed from carbon and hydrogen only was found to contain a 3.3% carbon. If 0.1 mole of this compound is weighing 15.6g, find (1) the empirical formula (2) the molecular formula.

w/w% C = 92.3%

w/w % H = 7.7%

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Subject Date No.

Assum you have 100 g of this compound ⇒ it contains 92.3 g C ⇒ moles C = $\frac{wt}{Awt}$ = $\frac{92.3\ g}{12\ g/mole}$ = 7.691 mole

and 7.7 g H ⇒ mole H = $\frac{wt}{Awt}$ = $\frac{7.7\ g}{1\ g/mole}$ = 7.7 mole

⇒ Empirical formula is CH.

Molecular formula = n * empirical formula

* 0.2 mole $\xrightarrow{\text{weight}}$ 15.6 g

1 mole ⟶ X

X = $\frac{15.6\ g \times 1\ mole}{0.2\ mole}$ = 78 g

n = $\frac{Mwt}{Emp.form.wt}$ = 1×12 + 1×1 = 13 g/mole

n = $\frac{78\ g/mole}{13\ g/mole}$ = 6

*molecular formula is C_6H_6

NOTEBOOK

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Ex.: A compound x formed from (C, H, O) was found to contain 68.8% C and 26.2% O. what is the molecular formula if It's molar mass is 122 g/mole?

w/w% H = 100% − (68.8% + 26.2%) = 5%.

Assoum you have 100 g of X ⇒ it contains

68.8 g C ⇒ moles C = $\frac{wt(C)}{Awt} = \frac{68.8\ g}{12\ g/mole} = 5.733$ mole

26.2 g O ⇒ moles O = $\frac{wt(O)}{Awt} = \frac{26.2\ g}{16.\ g/mole} = 1.637$ mole

5 g H ⇒ moles H = $\frac{wt(H)}{Awt(H)} = \frac{5\ g}{1\ g/mole} = 5$ mole

Devid by 1.637 : $\frac{5.733\ mole}{1.637\ mole} = 3.5 \rightarrow C$

2) $\frac{1.637\ mole}{1.637\ mole} = 1 \rightarrow O$

3) $\frac{5\ mole}{1.637\ mole} \simeq 3 \rightarrow H$

$2 \times (C_{3.5} H_3 O) \Rightarrow C_7 H_6 O_2$ (empirical formula)

molecular fomula = n * Emp. form

$n = \frac{122\ g/mole}{122\ g/mole} = 1$

molecular formula = $C_7 H_6 O_2$

$A_{1.33} B_2 \xrightarrow{\times 3} A_4 B_6$

$X_{1.246} Y_3 \xrightarrow{\times 4} X_5 Y_{12}$

NOTEBOOK

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Limiting Reactant :- It is the reactant which consumed first and on which the amount of product depends on.

$$1A + 2B \longrightarrow 1C + 3D$$

reactant — products

1 A needs → 2 mole B

2 A → X

X = 4

$$4\,Al_{(s)} + 3O_{2\,(g)} \longrightarrow 2Al_2O_{3\,(s)}$$

$$N_{2\,(g)} + 3H_{2\,(g)} \longrightarrow 2NH_{3\,(g)}$$

Example :- If 18 g Al was reacted with sufficient amount of oxygen O_2, find the amount of Al_2O_3 expected to be obtained?

Hint ⇒ (Awt Al = 27 g/mole, O = 16 g/mole)

$$4Al_{(s)} + 3O_{2\,(g)} \longrightarrow 2Al_2O_{3\,(s)}$$

Al: Limiting (s) — O_2: excess (g)

$$\text{moles Al} = \frac{\text{wt Al}}{\text{Awt Al}} = \frac{18\ g}{27\ g/mol} = 0.667 \text{ mole}$$

4 moles Al gives 2 moles Al_2O_3

0.667 → ? X

$$X = \frac{2 \times 0.667 \text{ mole}}{4 \text{ mole}} = 0.333 \text{ mole } Al_2O_3 \text{ produced}$$

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Mwt Al_2O_3 = $\frac{2 \times 27 \text{ g}}{54}$ mole + 3 × 16 g/mole = 102 g/mole

wt Al_2O_3 = moles × MWt = 0.333 mole * 102 g/mole = 34 g

Theoritical yield:- It is the amount of products got basing on calculations.

Actual yield:- It is the amount of products got from experiment in the lab

% yield = $\frac{\text{Actual yield}}{\text{Theoritical yield}}$ * 100%

Ex:- when 28.5 g of Mg was reacted with 32 g N_2 34.2 g of Mg_3N_2 was obtained. find the percentage yield?

Limiting → $3Mg_{(s)} + N_{2(g)}$ ← excess $\rightarrow Mg_3N_2$

(AWt Mg = 24.3 g/mole, N = 14 g/mole)

moles Mg = $\frac{\text{wt Mg}}{\text{AWt Mg}} = \frac{28.5 \text{ g}}{24.3 \text{ g/mole}}$ = 1.17 mole

⇒ moles N_2 = $\frac{\text{wt } N_2}{\text{MWt } N_2} = \frac{32 \text{ g}}{28 \text{ g/mole}}$ = 1.14 mole

3 Mg needs → 1 mole N_2

1.17 mole Mg → ? X

X = $\frac{1 \text{ mole } N_2 * 1.17 \text{ mole Mg}}{3 \text{ mole Mg}}$ = 0.39 mole N_2 required

Since what we have from N_2 (1.14 mole) is more than what we need ⇒ N_2 is excess ⇒ Mg is the limiting reactant

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3 moles Mg gives → 1 mole Mg_3N_2

1.17 " " → ? y

$$y = \frac{1 \text{ mole } Mg_3N_2 * 1.17 \text{ mole Mg}}{3 \text{ mole Mg}} = 0.39 \text{ mole } Mg_3N_2 \text{ produced}$$

wt of Mg_3N_2 = moles * MWt Mg_3N_2

$$= 0.39 \text{ moles} * \left(\frac{3*24.3}{72.9} + \frac{2*14}{28}\right) \text{ g/mole}$$

$= 39.4$ g theoritical yield

$$\% \text{ yield} = \frac{34.2}{39.4} \times 100\% = 86\%$$

② What is the mass of unreacted reactant?

unreacted moles N_2 = orginal moles − reacted moles

$0.75 \text{ mole} = 1.14 \text{ mole} - 0.39 \text{ mole}$

mass of unreacted N_2 = moles * MWt N_2

$= 0.75 \text{ mole} * 28 \text{ g} = 21 \text{ g } N_2$

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تفاعلات المحاليل

Reactions in Solutions

Solutions is formed from solute (مذاب) and solvent (مذيب).

Concentration of solutions represent the ratio between solute and solvent.

*Concentration can be expressed by:-

① w/w% = $\frac{\text{wt of solute}}{\text{wt of solution}} \times 100\%$ (unit = unitless)

② Molarity = $\frac{\text{moles of solute}}{\text{Volume solution (L)}}$ $\left(\text{unit-M} = \frac{\text{mole}}{\text{L}} = \text{mol} \cdot \text{L}^{-1}\right)$

③ Molality = $\frac{\text{moles of solute}}{\text{mass solvent (kg)}}$ $\left(\text{unitm} = \frac{\text{mole}}{\text{kg (solvent)}} = \text{mole} \cdot \text{kg}^{-1}\right)$

① w/w%; Example:- what is the concentration of a solution in m/m% resulted from dissolving 22.5g of $KMnO_4$ in 360 mL H_2O? (density = d(H_2O) = 1g/mL)

$$m/m\% = \frac{22.5\text{g}}{22.5 + 1\frac{\text{g}}{\text{mL}} * 360\text{mL}} = \frac{22.5\text{g}}{382.5\text{g}} = 5.88\%$$

Ex:- How much $CaCl_2$ exists in 245ml of 1.45% solution? ($d_{solution}$ = 1.25 g/mL)

mass of solution = d x v = 1.25 g/mL * 245 mL = 306.25g

wt of $CaCl_2$ = w/w% * mass solution = 1.45% * 306.25 = 4.54g

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② Molarity; Ex(1) what is the mass of $K_2Cr_2O_7$ is required to prepare 6.2 L of 0.5 solution?
(Awt K = 39.2 g/mole, Cr = 52 g/mole, O = 16 g/mole)

Mwt or fwt = $2\times39.2 + 2\times52 + 7\times16 = 294.4$ g/mole

$M = \frac{\text{moles}}{V_{(L)}} \Rightarrow \text{moles} = M \times V_{(L)} = 0.5 \frac{\text{mol}}{L} \times 6.2 L = 3.1$ mole

moles $K_2Cr_2O_7 = \frac{wt}{Fwt} \Rightarrow wt = \text{moles} \times fwt = 3.1 \times 294 = 918$ g

Ex(2); what is the molar concentration of a solution resulted from dissolving 16g KOH in 420 mL water?

Fwt = KOH = (39.2 + 16 + 1) = 56.2 g/mole

moles KOH = $\frac{wt}{fwt} = \frac{16}{56.2} = 0.28$ mole

volume solution = $\frac{420}{1000} = 0.42$ L

$m = \frac{0.284}{0.42} = 0.676$ M

③ Molality; Ex(1) what is the molal concentration of a solution resulted from dissolving 16.2 g of $CaCl_2$ in 830 mL H_2O? ($d_{H_2O} = 1$ g/m, Awt Ca = 40 g/mole, Cl = 35.5 g/mole)

Fwt $CaCl_2 = (2\times35.5 + 1\times40) = 111$ g/mole

moles $CaCl_2 = \frac{wt}{Fwt} = \frac{16.2}{111} = 0.145$ mole

$m = d \times V$

$= 1 \times 830 = 830$ g $= 0.83$ kg.

$m = \frac{\text{moles } CaCl_2}{\text{mass } H_2O} = \frac{0.145 \text{ mole}}{0.83 \text{ kg}} = 0.17$ m

1 L = 1000 ml = 10^3 cm^3, 1 m^3 = 10^3 L = 10^6 ml, 1 cm^3 = 1 ml = 10^{-3} L = 10^{-6} m^3

1 dm^3 = 1 L

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Subject Date 25th Nov No.

Ex(2): what is the mass $KMnO_4$ required to prepare 320mL of 0.6m? (Given that dsolution = 1.1g/mole)

mass of solution = dsolution * V = 1.1 * 320 = 352 g

Fwt = 158 g/mole

$$0.6m = \frac{\frac{x}{158}}{\frac{352-x}{1000}} \rightarrow 0.6 = \frac{x}{158} \times \frac{1000}{352-x} =$$

$$6\times10^{-4} = \frac{x}{55616\times10^{4} - 158x}$$

$$x = 33.366 - 0.0948x =$$

$$= \frac{1.0948x}{1.0948} = \frac{33.366}{1.0948} = 33.05\ g$$

mass of $KMnO_4$

*Interconversion from w/w% to molarity and vise versa

(1) from w/w% to molarity :-

Ex:- what is the molar concentration of 36% KNO_3 solution, given that density of the solution is 1.4g/mL?

Fwt KNO_3 = 101 g/m

molarity = $\frac{\text{moles } KNO_3}{V \text{ solution (L)}}$

* Assum you have 100 g solution $\Rightarrow$ it contains 36g KNO_3

$$\text{moles} = \frac{wt}{Fwt} = \frac{36g}{101} = 0.356 \text{ moles}$$

$$V = \frac{\text{mass}}{d} = \frac{100}{1.4} = 71.4\ mL = \frac{71.4}{1000\ mL} = 0.071\ L$$

$$\text{Molarity} = \frac{0.356 \text{ mole}}{0.071} = 5.01\ M$$

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Subject Date No.

ex(2) :- from molarity to mass percentage: what is the mass percentage (w/w% or m/m%) of 6.2 M H_2SO_4 solution? (Hint $d_{solution}$ = 1.3 g/mL) (MWt H_2SO_4 = 98 g/mole)

$$w/w\% = \frac{\text{wt of } H_2SO_4}{\text{wt of solution}} * 100\%$$

1000 mL ←

Assum you have 1L solution → it contains 6.2 mole H_2SO_4

m = moles * MWt = 6.2 * 98 g/mole = 607.6 g

$m = d * V$ = 1.3 g/mL * 1000 mL = 1300 g

$$w/w\% = \frac{607.6}{1300} * 100\% = 46.73\%$$

* concentration of ions:-

$$\overset{+1}{K_2}\overset{-2}{SO_4} \xrightarrow{H_2O} 2K^+ + \overset{-2}{SO_4}$$

Ex:- 0.1m KCl → $[K^+]$ = 0.1m

and $[Cl^-]$ = 0.1m.

ex:- 0.5m Na_3PO_4 → $[Na^+]$ = 1.5 M

$[PO_4^{3-}]$ = 0.5 M

ex:- which of the following solution has the highest Na^+ concentration?

(1) 0.2M Na_2SO_4

(2) 0.4 NaCl

(3) 0.3 Na_3PO_4

(4) 0.4 Na_3PO_4

(5) 0.5 Na_2CO_3

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Dilution of Solution :-

moles of solute beforedilution = moles of solute after dilution

$M_i * V_i = V_2 * M_2$

ex(1) :- How much water should be added to 200 ml. of 0.6 M K_2SO_4 solution to reduce its concentration to 0.3 solution?

$M_i V_i = M_f V_f$

$0.6M \times 200Ml = 0.3 \times V_2$

$V_2 = 400\ Ml$

Volume of water should be added $= V_2 - V_i = 200ml$

ex(2) :- How much water should be added to 600 mL of $KMnO_4$ solution to reduce its concentration to $\frac{1}{4}^{th}$?

sol $M_i V_i = M_f V_f$

$4 * M_i * 600ml = \frac{1}{4} * V_2 * M_i * 4 \Rightarrow V_2 = 2400ml$

$V_2 - V_i = 2400 - 600 = 1800ml$

ex(3) :- How much sduent should be remard to increase the conentration 4-folds?

$M_i V_i = M_2 V_2$

$M_i V_i = 4M_i V_2 \Rightarrow V_2 = \frac{1}{4} V_i$

$V_i - V_2 = V_i - \frac{1}{4} V_i = \frac{3}{4} V_i$

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Subject Date No.

ex(4) :- How much KOH should be added to 250 mL of 0.1 M solution to increas its concentration 3-folds?

$M_2 = 3 * M_1 = 3 * 0.1 = 0.3\,M$

moles $= M \times V$

$= 0.3 \times \frac{250}{1000} = 0.075$ mole

moles before addition $= M \times V$

$= 0.1 \times 0.25 = 0.025$ mole

moles should be added $= 0.075 - 0.025 = 0.05$ mole

$\text{moles} = \frac{wt}{fwt} \Rightarrow wt = \text{moles} * fwt$

$0.05 * 56 = 2.8\,g$

NOTEBOOK

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Reactions in solutions

Ex:- If 450 mL of 0.12 M $Ba(NO_3)_2$ was mixed with 240 mL of 0.26 M Na_3PO_4 solutions; 6 g of $Ba_3(PO_4)_2$ (precipitate راسب) was formed. Find (1) percentage yield (2) Find concentration of Ba^{2+} and Na^+ ions:-

excess — Limiting

$$2\,Na_3PO_{4(aq)} + 3\,Ba(NO_3)_{2(aq)} \longrightarrow Ba_3(PO_4)_{2(s)}\downarrow + 6\,NaNO_{3(aq)}$$

moles Na_3PO_4 = M * V = 0.26 mol/L * 0.24 L = 0.624 mole

moles $Ba(NO_3)_2$ = M * V = 0.12 mol/L * 0.45 L = 0.054 mol

2 moles Na_3PO_4 needs → 3 moles $Ba(NO_3)_2$

0.624 → ?

$x = \frac{3 * 0.624}{2} = 0.0936$ mole

since what we have from $Ba(NO_3)_2$ (0.054) is less than required ⇒ $Ba(NO_3)_2$ is the limiting reactant.

3 moles $Ba(NO_3)_2$ gives → 1 mole $Ba_3(PO_4)_2$

0.054 mole → T

$T = \frac{0.054}{3} = 0.018$ mole ⇒ * wt of $Ba_3(PO_4)_2$

= moles * fwt

= 0.018 * 601

= 10.8 g

Fwt $Ba_3(PO_4)_2$ = 601 g/mole

% yield = $\frac{\text{Actual}}{\text{Theor.}} \times 100\% = \frac{6}{10.8} * 100\% = 55.6\%$

$[Ba^{2+}]_{find} = 0$

$[Na^+]_{find} = \frac{\text{moles } Na^+}{V_{total}}$ = 3* moles Na_3PO_4 = 3 * 0.064 mole = 0.1872 mole

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Subject Date No.

$V_{total} = \frac{240 + 450}{1000} = 0.69\ L$

$[Na^+] = \frac{\text{moles } Na^+}{V_{total}} = \frac{0.1872\ \text{mole}}{0.69} = 0.2713\ M$

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Subject Date No.

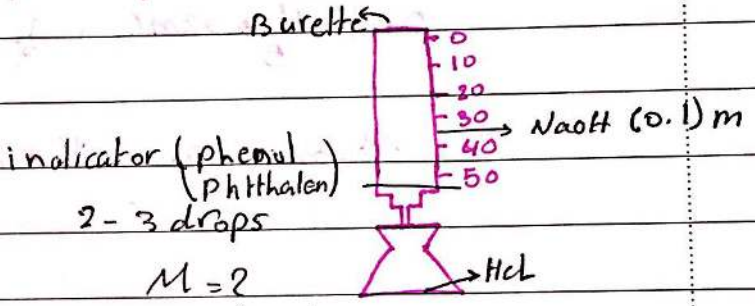
* APPLication on reactions in solutions :-

1) precipitation reaction تفاعلات الترسيب

2) Acid-Base rxn (r → reaction , n → Natralization rxn) تفاعلات حامض - قاعدة

* Titration (Acid-Base titration)

Titration :- it is an analytical technique in which a solution with known concentration is used to determind the concentration of another solution such that a complete rxn between them takes places.



* ex :- Upon addition of (0.1) M NaoH solution to (10mL) of HCl with unknown concentration, the Volume of Naoh consumed to neatralize Hcl solution was 24 ml . find the concentration of HcL solution?

$$HCl_{(aq)} + NaOH_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(l)}$$

نقطة التعادل

at end-point : moles HcL = moles NaoH

$$M_{HCl} \times V_{HCl} = M_{NaOH} \times V_{NaOH}$$

$$\frac{M_{HCl} * 10}{10} = \frac{0.1M * 24ml}{10}$$

$$M_{HCl} = 0.24 M$$

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Subject Date No.

ex(s) :- upon titration of 15 mL of KOH solution, 22 mL of 0.25M H_3PO_4 was required to neutralize the KOH. find the concentration of KOH solution?

$$3\,K^{+1}OH^{-1}_{(aq)} + H_3^{+1}PO_4^{-3} \longrightarrow K_3PO_{4(aq)} + 3H_2O_{(l)}$$

moles H_2PO_4 = $\frac{1}{3}$ moles KOH

$$M_{H_2PO_4} \times V_{H_2PO_4} = \frac{1}{3} M_{KOH} \times V_{KOH}$$

$$\frac{0.25M * 22mL}{5} = \frac{1}{3} * M_{KOH} * \frac{15mL}{5}$$

$$M_{KOH} = 0.05M * 22 = 1.1\ M$$

NOTEBOOK

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Subject Date 4th-Dec-2019 No.

الديناميكا الحرارية

Chapter 6 :- Thermodynamics :-

It's the branch of science dealing with energy changes accounpained to proceses.

Important Terms :-

* system :- It is the process under study.
* Surrounding (المحيط) :- every thing in the universe except the system.
* Universe = sys + surr

- systems can be derided into :-

(1) open system :- is the system in which both mass and heat is allowed to be transferd with Surr.

(2) closed system :- is the system in which the mass is not allowed to be transferd with surr, However the heat could be.

(3) Isolated system :- is the system in which Niether mass nor heat is allowed to be transferd with Surr.

(4) Isothermal system :- is the system in which the heat should be completly transferd with Surr ($\Delta T = 0$)

(5) Adiabatic system :- is the system in which the heat should not be transferd with Surr, However, the Mass could be ($\Delta T \neq 0$)

* Specific Heat (السعة الحرارية النوعية) (Sp) :- It is the heat required to raise the temperature (temp) of 1g of a substance 1°C.

$$\text{Unit} : \frac{J}{g.C^\circ} = J \cdot g^{-1} \cdot {}^\circ C^{-1}$$

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Subject Date No.

السعة الحرارية

* Heat Capacity (Cp). :- It is the heat required to raise the temp. of the substance 1°c.

$$unite = \frac{J}{°C} = J \cdot °C^{-1}$$

$$sp(H_2O) = 4.184 \frac{J}{g \cdot °C}$$

$$Cp = mass * Sp$$

* Heat Quantity (Q). :- It is the heat gained or losed

$$Q = Cp * \Delta T = m * Sp * \Delta T \rightarrow T_f - T_i$$

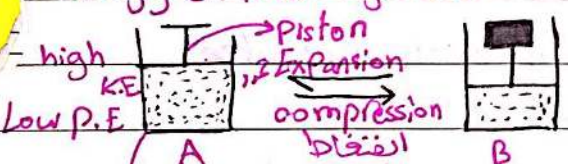
$$Q = g * \frac{J}{g \cdot °C} * °C$$

First Law of Thermodynamics :- The energy Neither created Nor destroyed, but it can be interconverted from one form to anther. i.e the energy is conserved

الطاقة الداخلية الحرارة

* Internal Energy :- It is the sum of Kinetic (K.E) and potential (P.E) energy of the system

طاقة الوضع طاقة الحركة



الإدخال الإخراج
معزول الحرارة متماثل الحرارة
isothermal system is high conductance.

$$E = K.E + P.E$$
$$E_B = (K.E + P.E)_B$$
$$E_A = (K.E + P.E)_A$$
$$\Delta E = E_B - E_A$$

for an idea gas and isothermal system; $\Delta E = 0$

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Subject Date No.

First Law :- $\Delta E = q_{heat} + w_{work}$

$Q_{H_2O} = -Q_{metal}$

$m.Sp._{H_2O} . \Delta T_{H_2O} = -(m.Sp. \Delta T)_{metal}$

ملاحظات :-

q is negative ⇒ system give up heat

q is positive ⇒ system absorb heat

w is positive ⇒ surr. do work on system

w is negative ⇒ system do work on surr.

ΔE is negative ⇒ exothermic process.

ΔE is positive ⇒ endothermic process.

* state function :- It is the function which is path independent. It depends only on initial and final (states points)

NOTEBOOK

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Subject Date No.

$\Delta E = q + w$ (work)

ΔE = change internal energy

$w = -p\,\Delta V$

pressure = force / Area $\Rightarrow p = \frac{F}{A}$, $w = F \cdot d$

$1\,J = 1\,N \cdot m$

$p\,a.m^3 = \frac{N}{m^2} * m^3$

$= 1\,J = 1\,p\,a.m^3$

Ex:- An ideal gas was expanded against a constant pressure of 3.2 atm from 6L to 15L. find the heat (q) accompanied to their process isothermal.

* $1\,atm = 1.1 \times 10^5\,pa = 101\,kpa$
* $1\,m^3 = 10^3\,L$
* $1\,atm = 760\,mmHg = 760\,torr$
* $1\,L = 10^{-3}\,m^3$

$w = -p\,\Delta V = -3.2\,atm * 1.01 \times 10^5 \frac{Pa}{atm} (15-6)L * 10^{-3}$

$= -2908.8\,J \approx -2.9\,KJ$

$q = +2.9$

Enthalpy (H):- It is the heat content under constant pressure. $H = E + PV$

H — E

$\Delta H = \Delta E + \Delta PV$

$\Delta H = \Delta E + p\,\Delta V + V\cancel{\Delta p}$ (under constant p)

$\Delta p = 0$

$\Delta H = \Delta E + P\Delta V$

Enthalpy = q_p (const. pressure)

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* standard Molar Enthalpy :- (ΔH°) :- It is the enthalpy of 1m of subs. under standard conditions (1atm / 25°c) STP

* Properties of ΔH :-

(1) It is a state function. (q, w) arenot state function, However ΔE is state function.

2) $\Delta H_{reverse\ rxn} = -\Delta H_{forward\ rxn}$

$C_{(s)} + O_{2\,(g)} \longrightarrow CO_{2\,(g)} \quad .... \quad \Delta H_{rxn} = -393.5 KJ$

$CO_2 \longrightarrow C_{(s)} + O_2 \quad ---- \quad \Delta H_{rxn} = +393.5 KJ$

3) It is an extensive property i.e.:- It depends on the amount of substance.

$5C + 5O_2 \longrightarrow 5CO_2 \rightarrow \Delta H^\circ_{rxn(2)} = 5 * \Delta H^\circ_{rxn(1)}$

Fussion $1H_2O \xrightarrow[\text{Freezing}]{\text{Fusion}} 1H_2O \quad ---- \quad \Delta H^\circ_{Fusion} = 2\ (KJ.mol^{-1})$

Vaporization $H_2O \underset{\text{condensation}}{\overset{\text{vap.}}{\rightleftharpoons}} H_2O \quad ---- \quad \Delta H^\circ_{vap} = +44\ (KJ.mol^{-1})$

sublimation $H_2O_{(s)} \rightleftharpoons H_2O \quad --- \quad \Delta H^\circ_{sub} = +60 (KJ.mol^{-1})$

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Subject Date No.

* How to measur ΔH_{rxn}? By one of the following methods:-

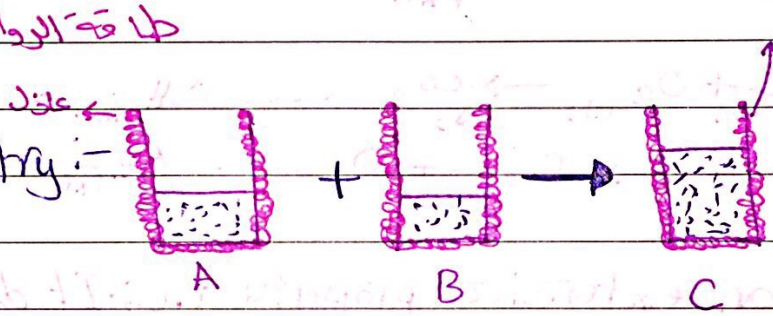
(1) Calorimetry → Cup. Calorimetry (for solution)
→ Bomb calorimetry (for gases)

(2) Hesse's Law قانون هس

(3) Enthalpy of formation حرارة التكوين

(4) Bond energy طاقة الروابط

* Cup – Calorimetry :-



ex:- Upon mixing 50 ml of 2 M HCl solution to 50 ml of 2 M NaOH solution is an isolated Cup-calorimetry, The temprature was raised from 18°C to 28°C Find ΔH° neutralization?

molar enthalpy $\Delta H = \frac{q}{n}$ → heat (KJ·mol^{-1}) Hint:- Sp final solution = 4.18 J·g^{-1}·c^{-1} (d solution = 1 g/mole)
→ no. of moles

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$

$q_{rxn} = q_{solution} - q_{calorimeter}$ (calorimeter = 0 الكأس) $= -mass(d*V) * Sp_{solution} * \Delta T_{solution}$

$= -(100*1)*4.18*(28-18)$

$= -4598\ J \cong -4.6\ J$

$\Delta H^\circ_{neutr.} = \frac{q_{rxn}}{n}$ → no of moles (H_2O)

moles HCl = M*V = 2*0.05 = 0.1 mol

$\Delta H^\circ = \frac{-4.6\ KJ}{0.1} = -46\ KJ \cdot mol^{-1}$

n = No. of moles of H_2O = No. of moles of HCl = No. of moles NaOH

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* Bomb-Celorimeter:- (we measure: $\Delta E = \Delta U$; the volume is constant

$q_{rxn} = \Delta E = \frac{q_v}{n}$ * $q_{rxn} = -q_{celorimeter} = (-q_{bomb} + q_{H_2O}) = -q_{bomb} - q_{H_2O}$

$= -C_{p_{calormeter}} * \Delta T = -C_{p_{bomb\,only}} * \Delta T - mass_{H_2O} * Sp(H_2O) * \Delta T$

** Example :- Upon combusting four grams of mythelhy hranzine (CH_6N_2) in a bomb calorimeter of 7.8 KJ/1C° heat capacity the temp. was raised from 25 C° to 39.5 C°. Find the molar heat of combustion of mythelhy hrazne (CH_6N_2 :- FWT = 46 g.mol^{-1})

$\Rightarrow q_{rxn} = -q = -C_p * \Delta T = -7.8\ Kg \cdot C^{-1} * (39.5 - 25.0)\ C^\circ = -113.1\ KJ$

$\Delta E = \frac{q_v}{n} = \frac{-113.1}{0.087} \sim -1301\ Kg \cdot mol^{-1}$

$\frac{wt}{mwt} = \frac{4}{46} = 0.087$ mole

[2] Hes's Law:- For the reaction resulted from individual steps, the enthaply of this reaction is equal to the sum of all enthalpies.

Example:- Given that:- ① $C_{(s)} + O_{2\,(g)} \longrightarrow CO_2$ .... $\Delta H_1^\circ = -393.5\ KJ.mol^{-1}$

② $CO_{(g)} + \frac{1}{2} O_2 \longrightarrow CO_2$ .... $\Delta H_2^\circ = -283\ KJ.mol^{-1}$

Find ΔH°_{rxn} : $C_{(s)} + \frac{1}{2} O_2 \longrightarrow CO$ .... $\Delta H^\circ_{rxn} = ?$

$C + \frac{1}{2} O_2 \longrightarrow CO_2$

$CO_2 \longrightarrow CO + \frac{1}{2} O_2$ ...

$C_{(s)} + \frac{1}{2} O_{2\,(g)} \longrightarrow CO_{(g)}$ .... $\Delta H^\circ_{rxn} = -393.5\ KJ \oplus 283\ KJ$

$= -101.5\ KJ$

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examples :- Given that :- ① $4C + 5H_2 + \frac{1}{2}O_2 \rightarrow C_4H_{10}O$ .... $\Delta H^\circ = -335 kJ$

② $4C + 4H_2 \rightarrow C_4H_8$ .... $\Delta H_2 = -17 kJ$

③ $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}$ .... $\Delta H_3 = -242 kJ$

Find ΔH°_{rxn} $= C_4H_{10}O \rightarrow C_4H_8 + H_2O_{(g)}$ .... $\Delta H^\circ = ?$

$-1 * (1) + (2) + (3)$

(1) $C_4H_{10}O \rightarrow 4C + 5H_2 + \frac{1}{2}O_2$ .... $\Delta H = 335 kJ$

(2) $4C + 4H_2 \rightarrow C_4H_8$ .... $\Delta H_2 = -17 kJ$

(3) $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ .... $\Delta H_3 = -242 kJ$

$\Rightarrow C_4H_{10}O \rightarrow C_4H_8 + H_2O$ .... $\Delta H^\circ_{rxn} = +335 - 17 - 242 = +76 kJ$

طاردة للتكوين

(3) Enthaply of formation: (ΔH°_f) It is the enthalpy accounpained to the formation of 1 mole of the substance from its original elements as they occur in nature.

Ex :- for phosphonic acid H_3PO_4

$\frac{3}{2}H_{2(g)} + P_{(s)} + 2O_{2(g)} \rightarrow 1H_3PO_4$ — يجب ان يكون واحد مول ΔH°_f مشان $\Delta H^\circ_{rxn} = \Delta H^\circ_f$

$\Delta H^\circ_{rxn} = \Delta H^\circ_f (H_3PO_4)$

Ex :- $\frac{3}{2}H_{2(g)} + \frac{1}{2}Br_{2(g)} + \frac{3}{2}O_{2(g)} \rightarrow H_3BrO_3$ $\Delta H^\circ_{rxn} \neq \Delta H^\circ_f (H_3BrO_3)$

لأن Br سائل وليس غاز

Ex :- $4Fe_{(s)} + 3O_{2(g)} \rightarrow \boxed{2}Fe_2O_3$ $\Delta H^\circ_{rxn} \neq \Delta H^\circ_f (Fe_2O_3)$

↳ for 2 moles

$\Delta H^\circ_{rxn} = \sum (n) \Delta H^\circ_f (products) - \sum (m) \Delta H^\circ_f (reactant)$

coefficient

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Ex.:- Given that $\Delta H_f^\circ(CO_2) = -393.5\ KJ.mole^{-1}$, $\Delta H_f^\circ(H_2O) = -286 KJ.mol^{-1}$
$\Delta H_f^\circ(C_2H_5OH) = -278\ KJ.mol^{-1}$ Find ΔH° combustion for ethanol?

$$C_2H_5OH + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O_{(l)}$$

$$\Delta H^\circ_{rxn} = [2 * \Delta H_f^\circ(CO_2) + 3 * \Delta H_f^\circ(H_2O)] - [1 * \Delta H_f^\circ(C_2H_5OH) + 3 * \Delta H_f^\circ(CO_2)]$$

$$= [2 * -393.5 + 3 * -286] - [1 * -278 + 3 * \Delta H_f^\circ(H_2O)]$$ ($\Delta H_f^\circ(H_2O) = 0$)

$$= -1367\ KJ.mol^{-1}$$

العشوائية

* ch.18 :- entropy (S°) :- Disorder

Factors affecting S° :-

(1) phase of the Substance $S^\circ_{gas} > S^\circ_{Liquid} > S^\circ_{solid}$

صلبة - سائلة - غازية

(2) number of atoms (directly propotional)

(3) No. of bonds (" ")

(4) MWt (directly propotional)

Gibb's Equation :- $G = H - TS$ → entropy

Gibb's Free energy — enthalpy — temp.

$$\Delta G = \Delta H - T\Delta S$$

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Subject Date No.

Third Law of Theromodynamies:-

It is impossible to reach a temp. below 0 K. at zero K, S = 0

$T(K) \geq 0$

Second Law of Thermodynamic deals with Spontainity التلقائية

* The spontainity can be determind from:-

(I) $\Delta G_{process}$

a- If ΔG = -Ve ⇒ the process is spontainity.

b- If ΔG = +Ve ⇒ the process is non-spontaneous.

c- If ΔG = 0 ⇒ the process is in equilibrium at constant

$\Delta G = \Delta H - T\Delta S$

T: temp. ΔS: entropy change

$\Delta S^\circ_{process\ rxn} = \sum nS^\circ(products) - \sum mS^\circ(reactant)$

2) $\Delta S_{universe}$

a- If $\Delta S_{univ.} > 0$ ⇒ the process is spontaneous.

b- If $\Delta S_{univ.} < 0$ ⇒ the process is non-spontaneous

c- If $\Delta S_{univ.} = 0$ ⇒ the process in equilibrium.

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ex:- consider the following reaction:-

$$C_{(s)} + \frac{1}{2} O_2 \longrightarrow CO_{(g)} \quad \cdots\cdots \quad \Delta H^\circ_{rxn} = -110.5\ KJ$$

Is this rxn is spontaneous at 300°c?

ΔG = -ve ⇒ because ΔS is +ve (increasing no. of moles of gases)

⇒ This rxn is spontaneous always

ex(2):- Consider the following rxn:-

$$3H_{2(g)} + N_{2(g)} \longrightarrow 2NH_{3(g)} \quad \cdots\cdots \quad \Delta H^\circ_{rxn} = -46\ KJ$$

Is this rxn is spontaneous at 25°C or not? why?

Hint $S^\circ_{NH_3} = 193\ J.mol^{-1}.K^{-1}$, $S^\circ_{N_2} = 198\ J.mol^{-1}.K^{-1}$, $S^\circ_{H_2} = 131\ J.mol^{-1}.K^{-1}$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta S^\circ = [2mole * S^\circ_{NH_3}] - [3 * S^\circ_{H_2} + 1 * S^\circ_{N_2}]$$

$$\Delta S = -199\ J.K^{-1}$$

للتحويل درجة الحرارة لكلفن نجمع 273

at 25°c ⇒ $K = °C + 273 = 25° + 273 = 298\ K$

$$\Delta G = \Delta H^\circ - T\Delta S^\circ$$

$$= -46\ KJ - (298 * -199)$$

$$\Delta G = +13.3\ KJ$$

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Subject Date No.

* $\Delta G = \Delta H - T\Delta S$ → always (+ve) third law.

* Third law of Thermodynamics :- it is impossible to mach a temp. belwo 0°K (-273C°) // 0°K ⟹ $S^\circ = 0$

	ΔG	ΔH	ΔS	Spontainity
1)	-ve	-ve	+ve	spontaneous at all temp.
2)	-ve	-ve	-ve	// only at law temp.
3)	-ve	+ve	+ve	// only at high temp.
4)	+ve	+ve	-ve	non-spontaneous at all temp.

$\Delta G^\circ_{rxn} = (3H_{2(g)} + N_{2(g)} \rightarrow 2NH_3 \quad \Delta G = +13.3\ KJ)$

$\Delta H^\circ = -46\ KJ$.

* In the perious example, at what temp. the rxn becomes spontaneous

$\Delta G = \Delta H_{rxn} - \Delta T\ \Delta S$

at T_{eq} ($\Delta G = 0$) → $0 = \Delta H - T_{eq} \cdot \Delta S$

* $T_{eq} = \frac{\Delta H^\circ}{\Delta S} = 231\ K$

∴ $T_{eq} = 231\ K = -42C^\circ$

This rxn is spontaneous belwo 231 K (or -42 C°)

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Subject Date No.

* ex :- Estimate the boiling point of water?

* ΔH°_{vap} . H_2O = +44 KJ

* $S^\circ_{H_2O_{(L)}} = 70\ J.mol^{-1}.K^{-1}$ / * $S^\circ_{H_2O_{(g)}} = 189\ J.mol^{-1}.K^{-1}$

$H_2O_{(L)} \rightleftharpoons H_2O_{(g)}$ $\Delta H^\circ_{vap} = +44\ KJ$

* $\Delta S_{vap} = S^\circ(H_2O)_{(g)} - S^\circ(H_2O)_{(L)} = (189-70)\ J.mol^{-1}.K^{-1}$

$= +119\ J.mol^{-1}.K^{-1}$

* Teq. = T. boiling $= \frac{+44000\ J}{119\ J.K^{-1}} = 369.7K$

$96.7C^\circ$

* $\Delta G^\circ f = \Delta H^\circ_f - {}^{298K}_{T} * \Delta S$ formation

* $\Delta G^\circ_{rxn} = \Sigma n\ \Delta G^\circ_f$ (prod.)

$- \Sigma m\ \Delta G^\circ_f$ (react)

* Max work = ΔG

* Efficiency = $\frac{work}{\Delta G}$ * 100%

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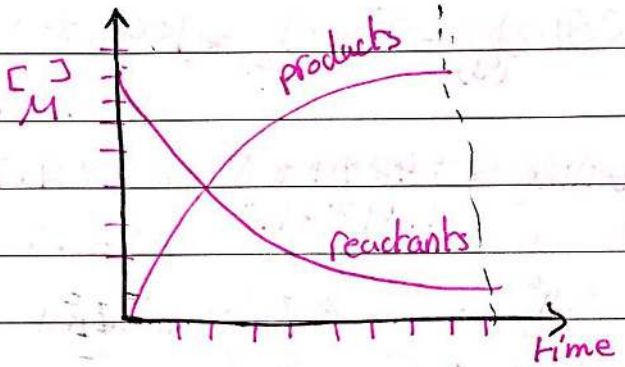
Subject Date No.

* chemical Kinetics . chapter 4 :- الكيمياء الحركية

- It is the branch of chemistry dealling with how fast the rxn proceeds

* Rate of rxn = $\frac{\Delta[\ ]}{\Delta t}$; Rate $= \frac{-[\text{reactant}]}{\Delta t} = \frac{+[\text{products}]}{\Delta t}$

Rate unit $M \cdot s^{-1}$



* ex : $1A + 2B \longrightarrow C + 3D$

Rate $= -\frac{\Delta[A]}{\Delta B} = \frac{-1}{2}\frac{\Delta[B]}{\Delta t} = \frac{+[\Delta C]}{\Delta t} = \frac{+1}{3}\frac{[\Delta D]}{\Delta t}$

* Ex :- Given the rxn : $3H_{2(g)} + N_{2(g)} \longrightarrow 2NH_{3(g)}$

- if the rate of precluction of ammoria (NH_3) is $0.24\ mol \cdot L^{-1} \cdot S^{-1}$, find the rate of consumption of hydrogen? ;

$\frac{1}{2}\frac{[\Delta NH_3]}{\Delta t} = \frac{-1}{3}\frac{\Delta[H_2O]}{\Delta t}$

$\Rightarrow \frac{[\Delta H_2]}{\Delta t} = \frac{3}{2} * 0.24\ mol^{-1} \cdot L^{-1} \cdot S^{-1} = -0.36\ mol \cdot L^{-1} \cdot S^{-1}$

↳ consumption

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Subject Date No.

Factors affecting the rate:-

(1) concentration of reactants (directly proportional)

(2) temperature (directly proportional)

(3) Nature of reactants

(4) Catalyst (increasing the rate of rxn)

$A + 2B \longrightarrow 3C + D$

Rate $\propto [A]^x$ x: the order of rxn with respect to A

Rate $\propto [B]^y$ y: " " " " " " B

* Rate $\propto [A]^x [B]^y \rightarrow$ Rate $= k [A]^x [B]^y$

k: const.

$x+y$ = overall order → الرتبة الكلية للتفاعل

* How to find x, y :-

To find x:- do two trials. Rate(1) $= k [A]^x [B]^y$

rate(2) $= k (2[A])^x [B]^y$

$$\frac{\text{Rate(2)}}{\text{Rate(1)}} = \frac{k * 2^x * [A]^x [B]^y}{k * [A]^x [B]^y} = 2^x$$

$2 = 2^x \quad x = 1$

if $\frac{\text{Rate(2)}}{\text{Rate(1)}} = 2 = 2^x \Rightarrow x = 1$

" $= 1 = 2^x \Rightarrow x = 0$

" $= 4 = 2^x \Rightarrow x = 2$

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if $\frac{Rate(3)}{Rate(1)} = 4 \Rightarrow 4 = 2^y \Rightarrow y = 2$

example :- consider the following rxn :-

$$2NO_{(g)} + Br_{2(g)} \longrightarrow 2NOBr_{(g)}$$

Trial	$[NO]^m$	$[Br_2]^n$	Rate ($M.s^{-1}$)
1	0.1	0.1	12
2	0.1	0.2	24
3	0.1	0.3	36
4	0.2	0.1	48
5	0.3	0.1	108

1) Find Rate Law

* $Rate = k[NO]^x[Br_2]^y$ to find x

$$\frac{Rate(5)}{Rate(1)} = \frac{k}{k} * \left[\frac{0.3}{0.1}\right]^x * \left[\frac{0.1}{0.1}\right]^y$$

$$\frac{108}{12} = 3^x \Rightarrow 9 = 3^x \quad x = 2$$

* $Rate = k[NO]^x[Br_2]^y$ to find y

$$\frac{Rate(2)}{Rate(1)} = \frac{k}{k} * \left[\frac{0.1}{0.1}\right]^2 * \left[\frac{0.2}{0.1}\right]^y$$

$$\frac{24}{12} = 2^y \Rightarrow y = 1$$

* $Rate = k[NO]^2[Br_2]^1$

K * unit :- $M^{-1}_{overall} \cdot S^{-1}$

2) Find K :-

$$12 = k[0.1]^2 * [0.1]$$

$$k = \frac{12}{0.001} = 12 \times 10^3 \; M^{-2}.S^{-1}$$

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* The relation between concentration and Time :-

(1) For first order $\ln \frac{[A]_0}{[A]_t} = kT$

$[A]_0$: initial conc. ≡ conc. at $T=0$ $[A]_T$

$[A]_t$: conc. after time (t)

* Half-Life :- It is the time required for the initial concentration to be reduced to its half value ($t_{\frac{1}{2}}$)

At half-life :-

$$\ln \frac{[A]_0}{\frac{1}{2}[A]_0} = k t_{\frac{1}{2}}$$

$$\ln 2 = k t_{\frac{1}{2}} \Rightarrow t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

Conclusion :- The half-life ($t_{\frac{1}{2}}$) in first order rxn is initial conc. $[A]_0$ independent لا يعتمد على التركيز الابتدائي

(2) second order - rxn $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$

at $t_{\frac{1}{2}} \Rightarrow \frac{1}{\frac{1}{2}[A]_0} - \frac{1}{[A]_0} = k t_{\frac{1}{2}} \Rightarrow \frac{1}{[A]_0} = k t_{\frac{1}{2}}$

$$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$$

(3) zero order rxn $[A]_0 - [A]_t = kt$

$t_{\frac{1}{2}}$:- $[A]_0 - \frac{1}{2}[A]_0 = k t_{\frac{1}{2}}$

$$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$$

NOTEBOOK

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Subject Date No.

Ex:- Consider the first order rxn:-

$2N_2O_{5(g)} \rightarrow 4NO_{(g)} + O_{2(g)}$ if $[N_2O_5]_0 = 1.00M$ and $[N_2O_5]_t = 3h$ is $1.21 \times 10^{-3} M$ find (1) half-life in (min) 2) N_2O_5 Concentration after 30 min?

* ① $t_{1/2} = \frac{0.693}{k}$, $\ln \frac{[A]_0}{[A]_t} = kt$

↳ after 3h $\rightarrow \ln \frac{1.0M}{1.21 \times 10^{-3} M} = k * (3 * 60 * 60)$

$k = 6.21 \times 10^{-4} \, s^{-1}$

* ② $\ln \frac{1.0M}{[N_2O_5]} = 6.21 \times 10^{-4} * 30 * 60$ e نقسم الطرفين على

$e \frac{1.0m}{[N_2O_5]_{t=20min}} = e^{1.11} = 3.1 \rightarrow [N_2O_5]_{t=30min} = \frac{1.0M}{3.1} = 0.32 M$

ex:- The decomposition of NOCl to NO and Cl_2 gives:-

$2NOCl_{(g)} \rightarrow 2NO_{(g)} + Cl_{2(g)}$

If $[NOCl_2] = 0.05 M$ and $k = 0.02 (L.mol^{-1}.s^{-1})$, find (1) half-life (2) The rate law (3) Percentage conversion after 1h?

* ① $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{6.21 \times 10^{-4} s^{-1}} = \frac{1115}{60} = 18.5 \, Min$

* from the unit, the rxn is 2nd order

$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{0.02 * 0.05} = 1 \times 10^{+3} \, s$

* ② Rate $= 0.02 M^{-1}.s^{-1} [NOCl]^2$

→

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Subject Date No.

*③ $[NOCl]_{t=1h}$; $\frac{1}{[NOCl]_{t=1h}} - \frac{1}{[NOCl]_0} = kt$

$\frac{1}{[NOCl]_{t=1h}} - \frac{1}{0.05} = (0.02\ M^{-1}s^{-1} * 1h * 60 \times 60)$

$= 72\ M^{-1}$

$[NOCl]_{t=1h} = 0.01086\ M$

$[NOCl]_{converted} = [NOCl]_0 - [NOCl]_{t=1hr}$

$= 0.05M - 0.01086\ M = 0.03914\ M$

$\%\ conversion = \frac{[NOCl]_{coverted}}{[NOCl]_0} \times 100\%$

$= \frac{0.03914\ M}{0.05\ M} \times 100\% = 78.28\%$

* The Effect of Temp. على التفاعل :-

$\frac{-G}{RT}$ -G :- Activation energy. It is the minimum energy required for the rxn to proceed

K : rate const. T :- Temp. (K)

$K \propto e$ R :- Gas const. $\Rightarrow (8.314\ \frac{J}{mol.K})$

$K = A\ e^{\frac{-Ea}{RT}}$ Arrehenous const. Equation.

$K_1 = A\ e^{\frac{-Ea}{RT_1}}$

$K_2 = A\ e^{\frac{-Ea}{RT_2}}$

$T_2 > T_1 \Rightarrow \frac{-Ea}{R} * (\frac{1}{T_2} - \frac{1}{T_1})$

$\ln \frac{K_2}{K_1} = \frac{A\ e^{\frac{-Ea}{RT_2}}}{A\ e^{\frac{-Ea}{RT_1}}}$

$\Rightarrow \ln \frac{K_2}{K_1} = (\frac{-Ea}{R} * \frac{1}{T_2} - \frac{1}{T_1})$

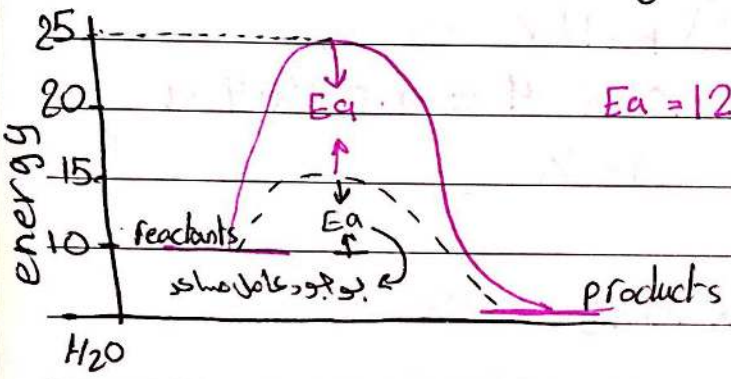
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ex:- Upon raising the temp. of a rxn from 25°C to 35°C, the rxn rate was increased by 15 folds, find the activation energy?

$$\frac{k_2}{k_1} = 15 \Rightarrow \ln 15 = \frac{-Ea}{8.314} * \left(\frac{1}{308K} - \frac{1}{298K}\right)$$

* The Effect of Catalyst:-



Ea = 12.5 kJ

* Oxidation . Reduction rxns:-

oxidation تأكسد:- Loss of e's

reduction اختزال:- gain of e's

$$A \longrightarrow A^{+3} + 3e \quad \text{(Oxidation)}$$

$$B + 2e^- \longrightarrow B^{2-} \quad \text{(Reduction)}$$

H → +1 ⇒ (إذا ارتبط مع فلز)

O → فوق بيركسيد (-1) H_2O_2

O → F_2O_2 +1 و F_2O +2 مع الفلور

* العجوزة الاولى عدد التأكسد (+1)

* " الثانية " (+2)

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